REPORT

Assessment Data Analysis

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Data Dictionary

1. **Invited_By_Email_Address** – Email address of the person who invited the candidate for the assessment.
2. **Appeared_On** – The date and time when the candidate appeared for the test.
3. **Candidate_ID** – Unique identifier assigned to the candidate.
4. **Masked Full Name** – Anonymized full name of the candidate.
5. **Masked Email** – Anonymized email address of the candidate.
6. **Candidate_Status** – Status of the candidate (e.g., Passed, Failed, In Progress, etc.).
7. **Test_Id** – Unique identifier for the test.
8. **Test_Name** – Name of the assessment test.
9. **Test_Status** – Status of the test (e.g., Completed, Terminated,Test Left).
10. **Test_Link_Name** – The label or identifier for the test instance.
11. **Test_Score** – Total score of the test.
12. **Candidate_Score** – Score obtained by the candidate.
13. **Test_Negative_Points** – Negative marking deducted from the candidate's score.
14. **Percentage** – Percentage score achieved by the candidate.
15. **Performance_Category** – Classification of performance (e.g., Beginner, Experienced)
16. **Total_Questions** – Total number of questions in the test.
17. **Test_Duration (minutes)** – Total duration of the test in minutes.
18. **Time_Taken (minutes)** – Time spent by the candidate to complete the test.
19. **Candidate_Feedback** – Feedback provided by the candidate about the test experience.
20. **Proctoring_Flag** – Proctoring status (e.g., Green - No violations, Red - Violations detected).
21. **Window_Violation** – Number of times the candidate switched windows during the test.
22. **Time_Violation (seconds)** – Total time (in seconds) spent violating test rules.
23. **Applicant_ID** – Unique identifier for the applicant (if linked to a job application).
24. **Percentile** – Candidate's percentile ranking among all test-takers.
25. **Section_Name** – Name of the test section (e.g., General Intelligence).
26. **Section_Score** – Maximum score for the section.
27. **Candidate_Section_Score** – Score obtained by the candidate in this section.
28. **Questions** – Total number of questions in the section.
29. **Not_Answered** – Number of questions not attempted by the candidate.
30. **Answered** – Number of questions attempted by the candidate.
31. **Correct** – Number of correctly answered questions.
32. **Wrong** – Number of incorrectly answered questions.

Data Preprocessing using Power Query and DAX

1. Null columns removed
   - Candidate Status
   - Application ID
2. Appeared On column split into:
   - Appeared On Time
   - Appeared On Date
3. Replaced null values in Proctoring Flag to “Unknown” and Feedback to “No feedback”.
4. “Candidate Summary” created for “Summary” Dashboard to calculate metrics grouped by “Candidate Name”.

```
Candidate_Summary =
SUMMARIZE(
    'Section Level Report',
    'Section Level Report'[Candidate_ID],
    'Section Level Report'[Masked Full Name],
    'Section Level Report'[Test_Name],
    'Section Level Report'[Candidate_Score],
    'Section Level Report'[Percentage],
    'Section Level Report'[Percentile],
    'Section Level Report'[Time_Taken(minutes)],
    'Section Level Report'[Performance_Category] ,
    'Section Level Report'[Test_Score]
)
```

Figure 1 DAX: Candidate Summary

5. Maximum percentile is calculated along with the candidate’s name for each test.

```
Max_Percentile_Candidate_Summary =
VAR SelectedTest =
    IF(
        HASONEVALUE('Candidate_Summary'[Test_Name]),
        SELECTEDVALUE('Candidate_Summary'[Test_Name]),
        BLANK()
    )

VAR MaxPercentile =
    CALCULATE(
        MAX('Candidate_Summary'[Percentile]),
        FILTER(
            'Candidate_Summary',
            ISBLANK(SelectedTest) || 'Candidate_Summary'[Test_Name] = SelectedTest
        )
    )

VAR CandidateName =
    CALCULATE(
        FIRSTNONBLANK('Candidate_Summary'[Masked Full Name], 1),
        FILTER(
            'Candidate_Summary',
            'Candidate_Summary'[Percentile] = MaxPercentile &&
            (ISBLANK(SelectedTest) || 'Candidate_Summary'[Test_Name] = SelectedTest)
        )
    )

RETURN
    CandidateName & " " & FORMAT(MaxPercentile, "0") & "%"
```

Figure 2 DAX: Maximum Percentile

6. Two measures are created to show “No Violations” for time and window violations.

PowerBi Report

Summary

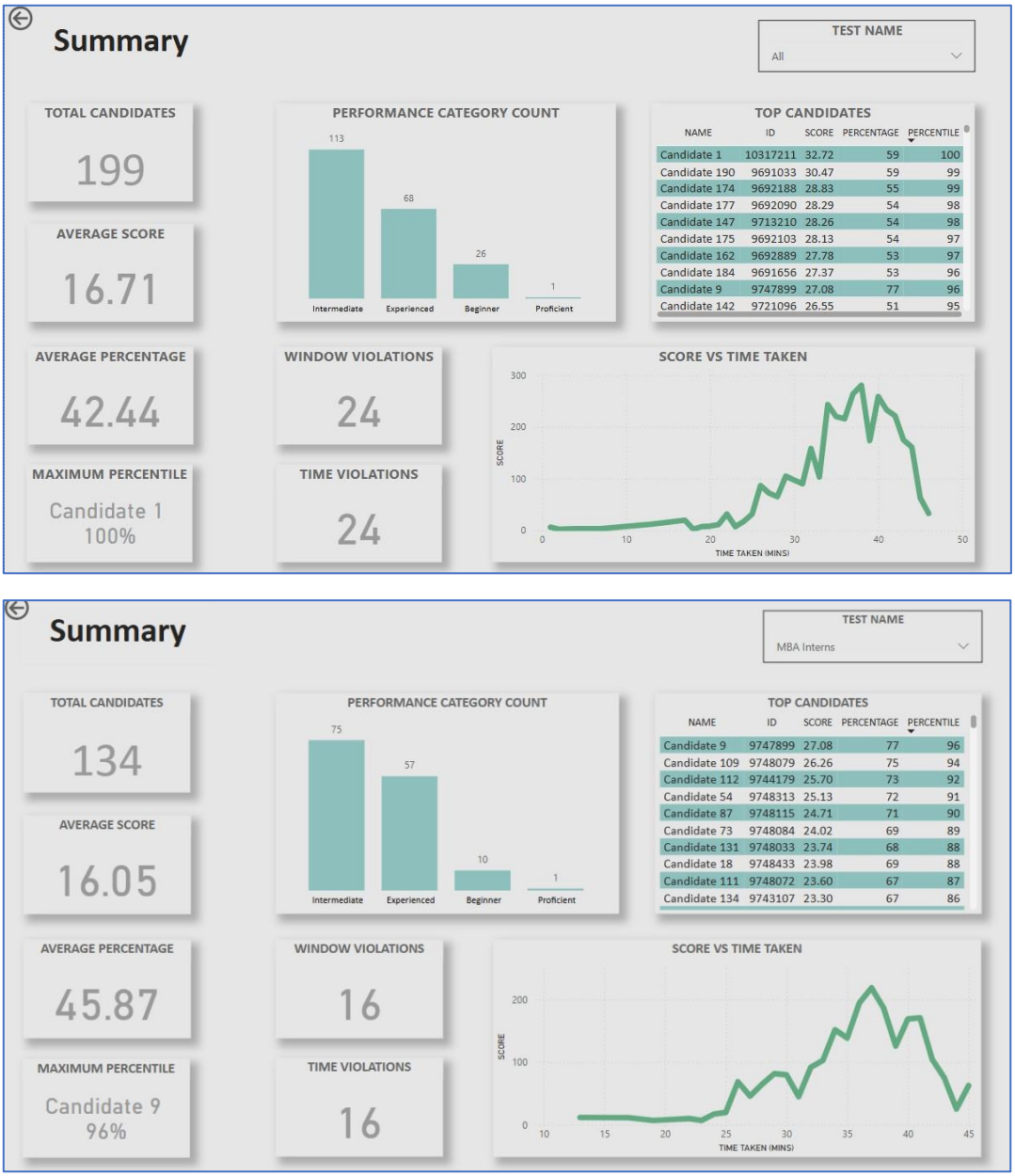
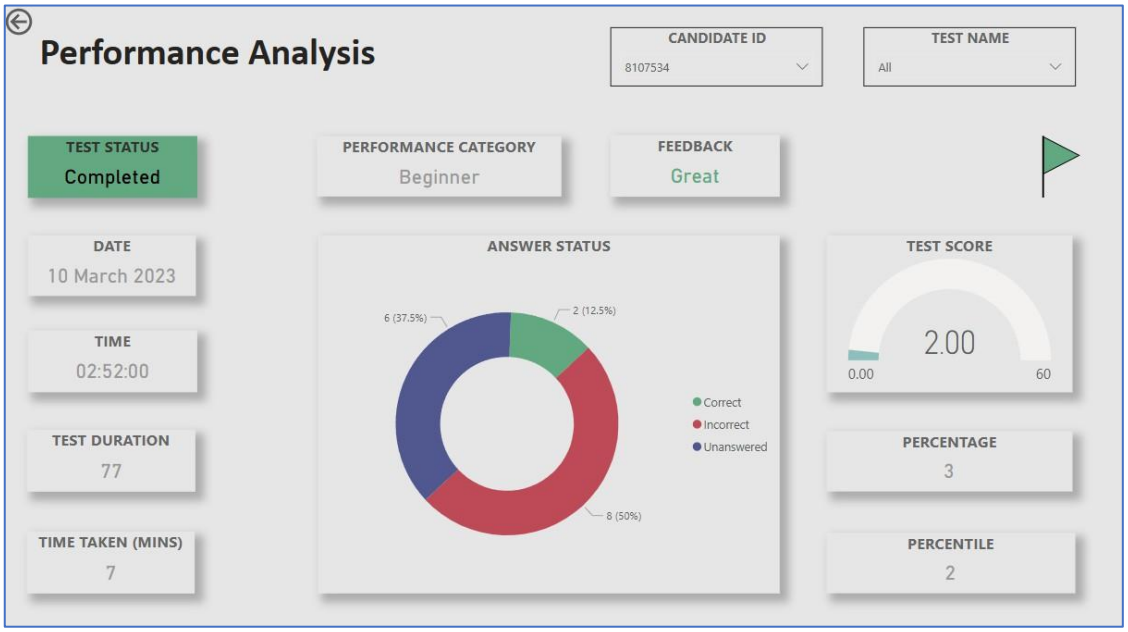


Figure 3 Summary Dashboard Snapshots

1. Slicer:
   - Test Name slicer to filter by different tests.
2. Visuals:
   - **Cards**: Total Candidates, Average Score, Average Percentage, Maximum Percentile
   - **Column Chart**: Performance category counts.
   - **Table**: Top candidates by percentile
   - **Line chart**: Relationship between time taken and test score.

Performance Analysis



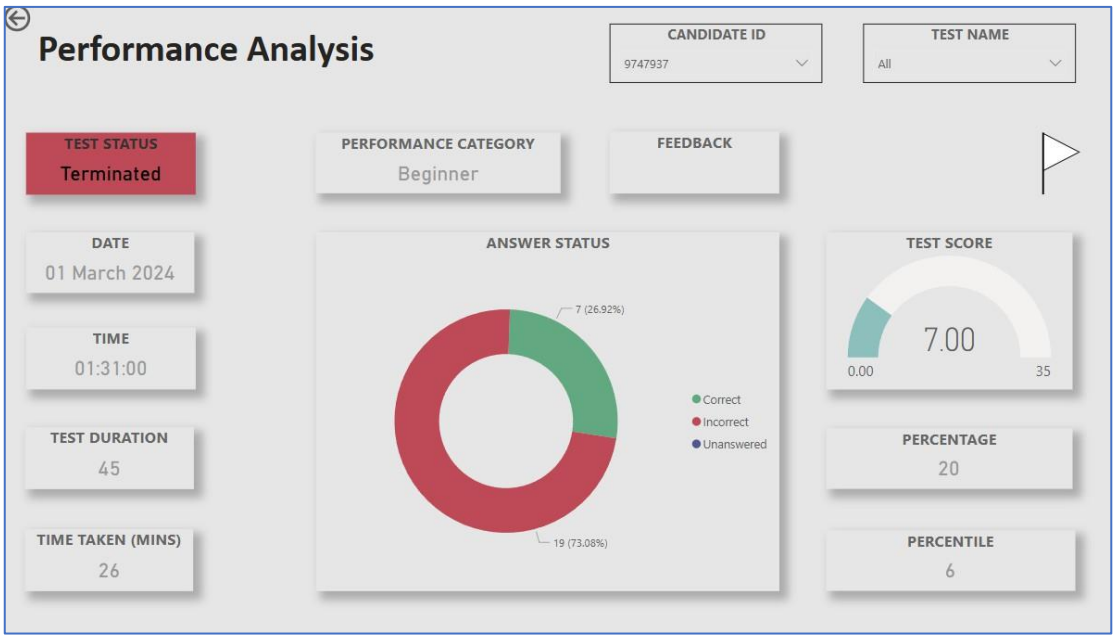


Figure 4 Performance Analysis Snapshots

1. Slicers:
   - **Candidate ID**: Filter by candidates ID.
   - **Test Name**: Filter by test name and test section.
2. Visuals:
   - **Cards**: Test Status (Different colours), Date, Time, Test Duration, Time taken by candidate, Performance category, Candidate feedback, Percentage, Percentile.
   - **Gauge**: Test score
   - **Donut chart**: Correct, Incorrect and Unanswered questions.
3. Proctor flag for violation status.

Test Compliance

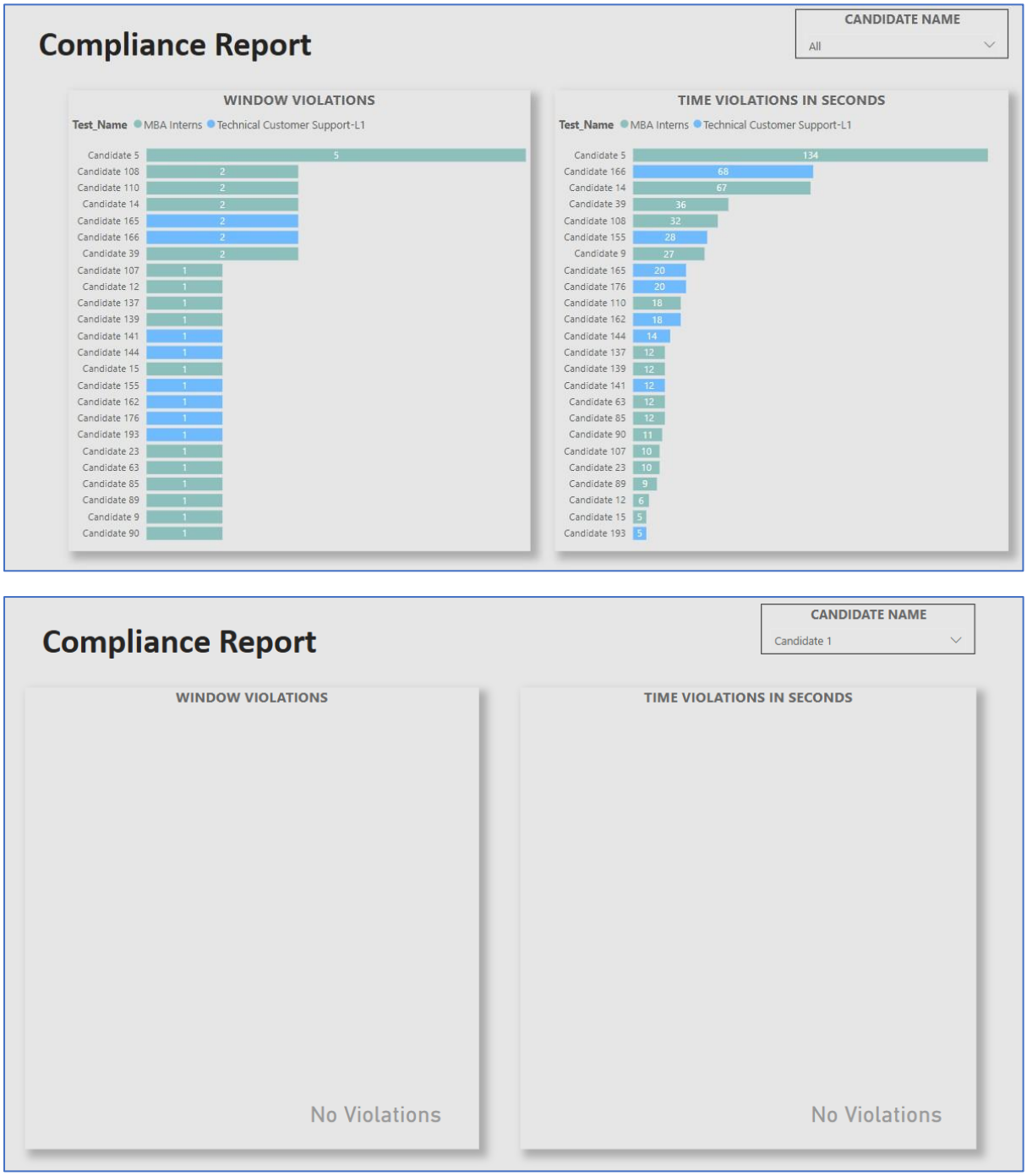


Figure 5 Compliance Report Snapshots

1. Slicer:
   - Candidate name slicer to filter by name.
2. Visuals:
   - **Card**: To show if there are no violations by a candidate.
   - **Bar Chart**: Window violations and time violations in seconds.

Descriptive Analysis

Summary

1. Overall, there are total 199 candidates and 7 Tests.
2. Average score is 16.71 and average percentage is 42.44.
3. Maximum percentile is 100% for Candidate 1.
4. Below is the table for detailed description:

	AWS	GalaxE Sol	iOS Developer	Manual Tester	MBA Intern	Senior Java Developer	Technical Customer Support
Total Candidates	1	1	1	1	134	1	60
Average Score	1	2	4	32.72	16.05	2	18.97
Average Percentage	2	6	10	59	45.87	3	36.45
Maximum Percentile	0	2	3	100	96	2	99

Table 1 Summary: Descriptive analysis

5. Performance Category Count:
   Beginner 26
   Intermediate 113
   Experienced 1
   Proficient 68
6. Total time violation and window violation are done by 24 candidates.

Performance Analysis

Let's take one candidate with Candidate ID 8107534:

1. Test Taken: Senior Java Developer
2. Test Status: Completed
3. Date:10 March 2023
4. Time: 02:52:00
5. Test Duration: 77 mins
6. Time Taken:7 mins
7. Correct-2, Incorrect-8, Unanswered-6
8. Test Score: 2/60
9. Percentage:3
10. Percentile:2
11. Performance Category: Beginner
12. Candidate Feedback: Great
13. Proctor Flag is Green.

Test Compliance

1. **Window Violations:**
   - There is total 24 candidates in Window violations.
   - Violations are in two tests: MBA Intern and Technical Customer Support
   - Most violations are by Candidate 5 which are 5.

2. **Time Violations:**
   - There is total 24 candidates in Time violations.
   - Violations are in two tests: MBA Intern and Technical Customer Support
   - Most violations are by Candidate 5 which are 134 seconds.

Diagnostic Analysis

Figure 6 Miscellaneous

1. **Decline in Average Scores Across Tests**
   - Candidates performed exceptionally well in the **Manual Tester 2022** test (Avg Score: 33), but the scores drop significantly in other tests, especially in **AWS Cloud Architect** (Avg Score: 1).
   - Possible reasons:
     a) **Variation in test difficulty**—some tests might be harder than others.
     b) **Candidate expertise**—more skilled candidates may have taken certain tests.
     c) **Test format differences**—some tests may have different scoring mechanisms.

2. **High Number of Unanswered Questions in Certain Tests**
   - **Technical Customer Support (69 unanswered) and MBA Interns (51 unanswered)** tests have the most skipped questions.
   - **AWS Cloud Architect (18 unanswered)** also shows a noticeable count.
   - Possible reasons:
   a) **Time constraints**—candidates may not have enough time to complete all questions.
   b) **Question complexity**—certain tests may have more difficult or ambiguous questions.
   c) **Test status impact**—terminated or abandoned tests might contribute to high unanswered counts.

3. **Impact of Proctoring Flags on Performance**
   - Candidates with **Green (No Violations)** had the highest average score (17.2), while **Red (Violations Detected)** and **Unknown** scored lower (14.6 and 14.1, respectively).
   - Possible reasons:
   a) **Cheating detection**— Candidates with violations may have engaged in unfair means but were still unsuccessful.
   b) **Stress factor**— Proctored tests may create anxiety, leading to lower performance.
   c) **Test environment differences**—technical issues or distractions could impact scores.

4. **Correlation Between Feedback and Scores**
   - Candidates who rated the test as **Great** scored **17.7**, while those who gave **Not Good** feedback had the lowest score (**9.5**).
   - Possible reasons:
   a) **Test design issues**—if many low-scoring candidates rated the test poorly, the test may need improvements.
   b) **Candidate mindset**—higher-performing candidates may naturally view the test more positively.